

Checkpoint 0: Choose Your Scenario

! Important

Due 10/11 (4 points). See [Project Central](#) for the full project overview.

Goal

Choose the public health scenario you will follow all quarter, read the published sources that come with it, and pull out the numbers you will use in Checkpoints 1–3.

You do **not** need to search for sources on your own. Each scenario below comes with a short **source packet**: the setting is fixed, and the papers you need are listed for you. Your job in this checkpoint is to read those sources, find the numbers, and describe them in your own words.

How the scenarios work

Each scenario card below has:

- **The setting:** who the population is and what happens to them. Please stay inside this setting all quarter.
- **The source packet:** 1–3 published sources. Every number you use must come from these sources or from a value the card gives you.
- **Course values:** a few numbers that the papers do not report in a usable form. The card gives you these directly. Use them as written.

i Words you might see in the papers

The papers use some terms we have not covered yet. You do not need to understand them fully for Checkpoint 0. Use this list to find the right number, and we will build each idea in class.

Term in the paper	What it means in plain words	Checkpoint used in
Prevalence	Out of everyone in the group, the fraction who have the condition	CP 1
Sensitivity	Out of the people who have the condition, the fraction who test positive	CP 1
Specificity	Out of the people who do not have the condition, the fraction who test negative	CP 1
False-positive rate	Out of the people who do not have the condition, the fraction who test positive ($1 - \text{specificity}$)	CP 1
Positive predictive value (PPV)	Out of the people who test positive, the fraction who have the condition. You will calculate this in CP 1 , so do not record it now.	CP 1
Abnormal interpretation rate (AIR)	Out of all screening exams, the fraction read as abnormal (the woman is called back)	CP 2
Cancer detection rate (CDR)	The number of cancers the screen found , per 1,000 exams. This is not the prevalence, because it leaves out the cancers the screen missed.	—
Secondary attack rate (SAR)	Out of the people exposed to a case (e.g., the people they live with), the fraction who get infected	CP 2
“Per 1,000” or “per 10,000”	A count out of 1,000 or 10,000 people. To turn it into a decimal, divide: 3 per 10,000 = $3 / 10,000 = 0.0003$.	CP 1, CP 2
Mean	The average value	CP 3
Median and interquartile range (IQR)	The middle value, and the range that holds the middle half of the values	CP 3
Standard deviation (SD)	How spread out the values are around the mean	CP 3
95% confidence interval (CI)	A range of plausible values around an estimate. Record only the main estimate , not the interval.	—

Scenario cards

Choose **one** of the three scenarios.

Scenario A

Community breast cancer screening program

The setting: A county health department runs a **mobile 3D mammography van** that brings free breast cancer screening to rural and underserved communities. The program screens women ages 40–74, following the current U.S. recommendation. A radiologist reads each exam as either normal or abnormal, and every woman with an abnormal read needs a follow-up diagnostic appointment. The program manager needs to know what an abnormal read means for the women served, how many follow-up appointments to reserve after each van day, and whether screening is finding cancers while they are still small.

Source packet

1. Lee CI, Abraham L, Miglioretti DL, et al. National performance benchmarks for screening digital breast tomosynthesis: Update from the Breast Cancer Surveillance Consortium. *Radiology*. 2023;307(4):e222499. [Free full text](#)

This one paper has everything you need. Look in the **Abstract** and the **Results** section. Use the numbers for **DBT** (digital breast tomosynthesis), not for digital mammography (DM).

What each checkpoint will look at

Checkpoint	What we model	Where it comes from
CP 1	The chance a woman has cancer, given an abnormal read	Sensitivity and specificity (paper); prevalence (course value below)
CP 2	The number of abnormal reads (follow-up appointments needed) out of 40 screens on one van day	Abnormal interpretation rate (paper)
CP 3	The size of a tumor found by screening, in millimeters	Median and IQR (paper); mean and SD (course values below)

Course values

- **Prevalence:** 6.6 cancers per 1,000 women screened. (The paper reports the *cancer detection rate*, which only counts the cancers that the screen found. We divided it by the sensitivity to include the cancers the screen missed.)
- **Screens per van day:** 40.

- **Tumor size:** mean of about 14 mm, SD of about 9 mm. (The paper reports a median and IQR. These values match that median and IQR for a right-skewed shape.)

💡 Finding your numbers (Scenario A)

Row in your Checkpoint 0 table	What it is called in the source
CP 1: prevalence	Course value on this card
CP 1: sensitivity	“Sensitivity” (DBT)
CP 1: specificity	“Specificity” (DBT)
CP 2: chance of a “yes”	“Abnormal interpretation rate” or “AIR” (DBT). A “yes” is a screen read as abnormal. Do not use the cancer detection rate (CDR) or PPV1.
CP 2: group size	Course value on this card
CP 3: mean and SD	Course values on this card. The paper reports a median and IQR for tumor size, which you may list too.

The abstract reports each measure for DBT and for DM side by side, so double-check that you are copying the DBT number.

Scenario B

County COVID-19 testing and contact tracing

The setting: A county health department gives out free rapid at-home antigen tests so that residents **without symptoms** can screen themselves, for example before visiting an older relative. When a resident tests positive and is confirmed as a case, the department’s contact tracers follow up with the people they live with. The department needs to know how much to trust a positive screening test, how many household members of each case are likely to be infected (to plan test kits and isolation support), and how long household contacts should watch for symptoms.

Source packet

1. Cochrane. How accurate are rapid antigen tests for diagnosing COVID-19? Plain language summary of Dinnes J, et al. *Cochrane Database of Systematic Reviews*, 2025 update (CD013705). [Summary page](#) · [Full review](#)
2. Madewell ZJ, Yang Y, Longini IM Jr, Halloran ME, Dean NE. Household secondary attack rates of SARS-CoV-2 by variant and vaccination status: An updated systematic review and meta-analysis. *JAMA Network Open*. 2022;5(4):e229317. [Free full text](#)

3. Del Águila-Mejía J, et al. Secondary attack rate, transmission and incubation periods, and serial interval of SARS-CoV-2 Omicron variant, Spain. *Emerging Infectious Diseases*. 2022;28(6). [Free full text](#)

For source 1, use the **overall** numbers for people **without symptoms**. For source 2, use the number for the **Omicron** variant in the **Abstract**. For source 3, use the **Omicron** incubation period in **Table 2**.

What each checkpoint will look at

Checkpoint	What we model	Where it comes from
CP 1	The chance a resident is infected, given a positive rapid test	Sensitivity and specificity (source 1); prevalence (course value below)
CP 2	The number of household contacts of one case who get infected	Household secondary attack rate for Omicron (source 2)
CP 3	The incubation period: days from infection until symptoms start	Mean and SD of the incubation period for Omicron (source 3)

Course values

- **Prevalence:** 0.5% (5 out of every 1,000 residents without symptoms who test). This is the example prevalence the Cochrane summary uses for general screening.
- **Household contacts per case:** 3 (a household of 4 people, with 1 case).

💡 Finding your numbers (Scenario B)

Row in your Checkpoint 0 table	What it is called in the source
CP 1: prevalence	Course value on this card
CP 1: sensitivity	“Average sensitivity” (source 1). Use the average across all evaluations, not the numbers for subgroups such as “general screening” or “suspected exposure”.
CP 1: specificity	“Average specificity” (source 1), also across all evaluations
CP 2: chance of a “yes”	“Household secondary attack rate” or “household SAR” for Omicron (source 2). A “yes” is a household contact who gets infected.
CP 2: group size	Course value on this card

CP 3: mean and SD

The “Incubation period” row for **Omicron** in Table 2 (source 3). The table lists each value as “Mean (SD)”, so the number in parentheses is the SD. Use the overall row, not the rows for vaccinated or unvaccinated index case-patients.

For source 1, use the numbers on the linked **2025** summary page. Older versions of this review report different numbers.

Scenario C

State newborn heart screening program

The setting: A state health department runs a **newborn screening program** that requires every baby born in the state to get pulse oximetry screening before going home. Pulse oximetry is a painless test that measures the oxygen level in the blood. It looks for critical congenital heart defects (CCHD), which are serious heart problems that need treatment soon after birth. The state has about 40,000 births a year. The program needs to know what a positive screen means for families, how many babies with CCHD to expect each year (to plan follow-up with pediatric heart specialists), and what typical birth weight looks like across the state’s newborns.

Source packet

1. Cochrane. Pulse oximetry for diagnosis of critical congenital heart defects. Plain language summary of Plana MN, Zamora J, Suresh G, et al. *Cochrane Database of Systematic Reviews*. 2018;3:CD011912. [Summary page](#) · [Full review](#)
2. Donahue SMA, Kleinman KP, Gillman MW, Oken E. Trends in birth weight and gestational length among singleton term births in the United States: 1990–2005. *Obstetrics & Gynecology*. 2010;115(2 Pt 1):357–364. [Free full text](#)

For source 1, look for the numbers “per 10,000 newborns” and the **false-positive rate**. Use the false-positive rate instead of the specificity, because the specificity (99.9%) is rounded too much to be useful. For source 2, use the mean and SD for **2005**.

What each checkpoint will look at

Checkpoint	What we model	Where it comes from
CP 1	The chance a baby has CCHD, given a positive screen	Prevalence, sensitivity, and false-positive rate (source 1)
CP 2	The number of babies born with CCHD in the state in one year	Prevalence (source 1); births per year (course value below)
CP 3	Birth weight, in grams	Mean and SD for 2005 (source 2)

Course values

- **Births per year in the state:** 40,000.

💡 Finding your numbers (Scenario C)

Row in your Checkpoint 0 table	What it is called in the source
CP 1: prevalence	The number of newborns out of 10,000 who have CCHD (source 1). Convert it to a decimal (see “per 10,000” in the word list above).
CP 1: sensitivity	“Sensitivity” (source 1)
CP 1: false-positive rate	“False-positive rate” (source 1). Write it as $P(T^+ D^c)$, not as a specificity.
CP 2: chance of a “yes”	The same prevalence as your CP 1 row. A “yes” is a baby born with CCHD.
CP 2: group size	Course value on this card
CP 3: mean and SD	Mean and SD of birth weight for 2005 (source 2), in grams

🔥 Want to propose your own scenario?

You may propose your own scenario **only with my approval**. Email me by 10/05 with:

1. the setting (who, where, and what test or classification),
2. **one to three published sources** that give you the same kinds of numbers as the scenario cards above: a prevalence, a sensitivity, and a specificity for CP 1; the chance of a “yes” for a count in CP 2; and a mean (and SD, if available) for a time or measurement in CP 3.

If you do not have my approval by the due date, please choose one of the three scenarios above.

Tasks

1. Team and scenario

- State whether you are working alone or with a partner (and your partner’s name).
- State which scenario you chose (A, B, or C).

- In 3–5 sentences, and **in your own words**, describe the scenario: Who is the population? What is being tested, counted, or measured? What decision does the public health program need to make, and why does it matter for the population it serves?

2. Find your numbers in the sources

Fill in the table below for your scenario. For each row:

- **Value:** Write the number from the source (or the course value from your scenario card).
- **Where I found it:** Name the source (1, 2, or 3) and the section, table, or sentence where the number appears (for example, “Source 1, Abstract, Results”). For a number from your scenario card, write “course value”.
- The **Finding your numbers** box on your scenario card tells you what each number is called in the source.
- **Probability statement:** For the three Checkpoint 1 rows only, write the value as a probability. Define your events first, for example, D = has the condition and T^+ = tests positive. Then write, for example, $P(T^+ | D) = 0.874$.

CP	What you are looking for (in plain words)	Value	Where I found it	Probability statement (CP 1 only)
1	Out of everyone screened or tested, the fraction who have the condition			
1	Out of the people who have the condition, the fraction who test positive			
1	Out of the people who do not have the condition, the fraction who test negative (A and B) or test positive (C)			
2	The chance of a “yes” for one screen (A: read as abnormal), one household contact (B: gets infected), or one baby (C: born with CCHD)			—
2	How many screens, household contacts, or babies are in the group			—

CP	What you are looking for (in plain words)	Value	Where I found it	Probability statement (CP 1 only)
3	The mean (average) of the time or measurement, with units			—
3	The SD of the time or measurement, with units			—

⚠ Record only what the source says

- Copy the main estimate only. Do not include the 95% confidence interval.
- Do not calculate anything new yet. For example, do not calculate the chance that someone with a positive test has the condition. That is the job of Checkpoint 1.
- Convert percents and “per 1,000” or “per 10,000” values to decimals in the probability statement column (e.g., 87.4% becomes 0.874, and 3 per 10,000 becomes 0.0003).

3. Your guiding questions

Write one question for each checkpoint that your analysis will answer. You may use the examples below for your scenario, or change them (for example, a different cutoff).

Scenario A (community breast cancer screening program)

- **CP 1:** If a woman screened on the van gets an abnormal read, what is the probability that she has breast cancer?
- **CP 2:** If the program reserves 6 follow-up appointments after each van day of 40 screens, what is the probability that it needs more than 6?
- **CP 3:** What is the probability that the average size of the next 30 tumors the program finds is more than 15 mm?

Scenario B (county COVID-19 testing and contact tracing)

- **CP 1:** If a resident without symptoms gets a positive rapid test, what is the probability that they are infected?
- **CP 2:** For a case who lives with 3 other people, what is the probability that at least 2 of them also get infected?
- **CP 3:** What is the probability that the average incubation period across the county’s next 50 cases is more than 3.5 days?

Scenario C (state newborn heart screening program)

- **CP 1:** If a newborn has a positive pulse oximetry screen, what is the probability that the baby has CCHD?
- **CP 2:** With 40,000 births a year, what is the probability that the state has 30 or more babies with CCHD in one year?
- **CP 3:** What is the probability that the average birth weight of 100 babies born in one county in a month is less than 3,300 grams?

These can change later. This step is to make sure your scenario works for all three checkpoints.

Submission

Submit a rendered HTML file (and your .qmd) with the three sections above. Make sure the qmd and html are named `lastname_firstname_checkpoint_00.qmd` and `lastname_firstname_checkpoint_00.html` or `lastname1_lastname2_checkpoint_00.qmd` and `lastname1_lastname2_checkpoint_00.html`. One submission per team. Either person in the pair can submit it.

Rubric (4 points)

Criterion	1 point	0 points
Scenario is chosen (or approved) and described in your own words	Yes	Missing, or not in your own words
Input table is complete, with the correct numbers from the sources or course values	Yes	Incomplete or incorrect
Every value says where it was found, and the three CP 1 rows have correct probability statements	Yes	Missing locations or incorrect statements
One guiding question for each checkpoint	Yes	Missing